Nikhil Kumar

(515) 817-9350 | nkumar88@iastate.edu | linkedin.com/in/nikhilkumar631 | nikhil-kumar-gilt.vercel.app | U.S. Citizen

Education

Iowa State University GPA: 3.92/4.00

B.S. Computer Engineering, Minor in Data Science *B.S. Expected May 2027 | Planned M.S. May 2028*

Experience

Boeing Research Fellow Aug 2026 – Present

Iowa State University *Ames, IA*

- Developing methods to validate assertions for Verilog and SystemVerilog RTL designs, emphasizing functional correctness and expected hardware behavior.
- Writing SystemVerilog Assertions and applying formal verification workflows with Yosys, SymbiYosys, and OSS CAD Suite.
- Investigating techniques for detecting incorrect or ineffective assertions in RTL designs, including hardware generated using large language models.

R&D Software Engineering Intern May 2026 – Aug 2026

Corteva Agriscience *Johnston, IA*

- Developed a production full-stack web application using Angular, TypeScript, C#, .NET 10, and SQL to support internal operational workflows.
- Built responsive desktop and mobile interfaces and integrated them with REST APIs and database-backed services.
- Worked directly with an internal customer to gather requirements, incorporate feedback, resolve UAT defects, and deliver an application now used in day-to-day operations.

ARA Research Program – REU Jan 2025 – Sep 2025

Iowa State University *Ames, IA*

- Built a CKAN-based data commons that provided researchers with centralized access to experimental datasets and documentation.
- Developed and containerized a Flask-based Moshell API for gNB components, creating reproducible Docker environments for wireless experimentation.
- Analyzed LoRa performance under varying weather conditions using Python-based correlation and regression across throughput, RSSI, SNR, and latency.

Projects

Vent Programming Language | *Go* Jan 2026 – Sep 2026

- Built a general-purpose interpreted programming language in a two-person team, implementing a modular lexer, parser, abstract syntax tree, and interpreter.
- Implemented integers, floats, characters, arrays, type inference, user-defined functions, control flow, and a reference-counted runtime for managing dynamically allocated objects.

Cynance – Student Budgeting Application | *Java, Spring Boot, MySQL, Android* Jan 2025 – May 2025

- Collaborated in a four-person team to build a full-stack Android application for expense tracking, budget management, and AI-assisted financial guidance.
- Developed Spring Boot REST APIs, MySQL data models, frontend features, and end-to-end integrations, earning Best Project and individual Best Coder awards.

Leadership

Treasurer Aug 2025 – May 2026

Indian Students Association *Iowa State University*

- Managed a $13,000 annual budget, overseeing reimbursements, vendor payments, and financial tracking for events.

Technical Skills

Languages: Java, Python, C, C++, C#, Go, SQL, TypeScript, JavaScript, Bash, Verilog, SystemVerilog

Frameworks: Angular, .NET, Spring Boot, React, React Native, Node.js, Flask, Android, JUnit

Data & Databases: PostgreSQL, MySQL, MongoDB, pandas, NumPy, scikit-learn, matplotlib

Tools: Azure DevOps, Docker, Git, GitHub, GitLab, Linux, Yosys, SymbiYosys, OSS CAD Suite